

4001CMD: Mathematical Skills for Computing Professionals

Week 1: Introduction, Sets and Basic Logic

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Semester 1, 2025-26

► What this module is about?

Discrete Mathematics !

► What is **discrete mathematics**?

- Mathematically, it's counting, combinatorics (ways of combining things), logic, basic set theory, basic number theory... other things.
 - It is NOT calculus, geometry, physics, mechanics, or anything involving continuous stuff.
 - Practically, it is the basis for how computers "think".
- ## ► Broadly speaking, we'll be looking at two big mathematical topics this semester and their applications to computer science:
- Number theory (Application: Cryptography)
 - Graph theory (Applications: Algorithms, Data compression, binary encoding, and error correction)

Provisional Weekly Topics

- ➊ Sets, Propositions and Basic Logic
- ➋ Symbolic Logic cont., Rules of Inference, Proofs.
- ➌ Axiomatic Proof and Induction
- ➍ Functions and Cardinality
- ➎ Combinatorics
- ➏ Number Theory

Online Core Assignment (pass/fail, starts in Week 7, 3 attempts (Weeks 7,8,9))

- ➐ Modular Arithmetic & RSA
- ➑ Algorithms & Graph Theory
- ➒ Graph Traversal & Searching Algorithms, Trees
- ➓ Heapsort and Huffman encoding
- ➔ Humming codes and bin packing

Applied Core Assignment (module mark ≥ 40) starts: Week 9, due: end of Week 12

1 Introduction

1.1 Sets

- 2 Sets – Definitions
- 3 Sets Notations
- 4 Elements of Sets
- 5 Sets of numbers
- 6 Subsets

- 7 Equal sets
- 8 Empty set
- 9 Union, Intersection, Difference,
Symmetric Difference
- 10 Venn diagrams
- 11 Sets' Summary



- ▶ We now turn to the 1st class of fundamental objects: Sets
- ▶ **Set theory is the foundation of modern mathematics.**
- ▶ Sets are fundamental building blocks used to describe collections of objects.
- ▶ They provide a formal way to structure and organise data.
- ▶ Many concepts in computer science, from data structures to databases, are built upon the principles of set theory.
- ▶ In this lecture, we will explore the basic definitions, operations, and properties of sets.

- ▶ A set is (roughly speaking) a **collection of things**. They can be numbers, words, symbols, poems, anything.
- ▶ Sets can be **finite**, or **infinite**.
- ▶ Examples:
 - $A = \{1, 2, 5\}$ (the set containing the numbers 1, 2, and 5)
 - $B = \{\text{John, Paul, George, Ringo}\}$
 - $X = \{a, b, c\}$ (the set containing a , b , and c — an abstract set with three elements)
 - $Y = \{n : n \text{ is a multiple of } 11\}$
- ▶ Here “:” reads as “such that” (alternative symbol is “|”).

► Notice a few things:

- A set is always denoted with the braces $\{\dots\}$, and is usually given a name, like A , B , X , or Y etc.
- A **finite** set is written as a list, separated by commas:



Roster notation:

$$A = \{1, 2, 5\}$$

- An **infinite** set cannot be listed in this way. So we describe the members of the set with a general pattern:



Set-builder notation:

$$Y = \{n : n \text{ is a multiple of } 11\}$$

- You would read the above as: “ Y is the set of all numbers n such that n is a multiple of 11.”
- The symbol “:” is read as “such that”. On the left is the type of object in the set; on the right is the property that these objects must satisfy.
- Sometimes the symbol “|” is used instead of “:”, like so:

$$Y = \{n | n \text{ is a multiple of } 11\}$$

- ▶ Objects in a set are called elements or members of the set.
- ▶ Example: Consider the set $A = \{1, 2, 5\}$. Then 2 is an element of A .
- ▶ The symbol “ $\in$ ” is used to show that something is an element of a set. Like so:

$$2 \in A \quad \longleftrightarrow \quad 2 \text{ is an element of } A \quad \longleftrightarrow \quad 2 \text{ is in } A.$$

- ▶ What about the set

$$Y = \{n : n \text{ is a multiple of } 11\} ?$$

- $22 \in Y$, $55 \in Y$, and $121 \in Y$ because 22, 55, and 121 are all multiples of 11. They are all elements of Y .
- ▶ But 23 is not a multiple of 11, and so is not an element of Y . We express this with the symbol “ $\notin$ ”. In this case, $23 \notin Y$, or:

$$23 \notin \{n : n \text{ is a multiple of } 11\}.$$

Another way to write this is $23 \in Y^c$, where Y^c is **the complement of Y** — everything not in Y .

- ▶ Some sets of numbers show up so often that we give them special names. There are four in particular that we'll look at from time to time:
 - $\mathbb{N} = \{1, 2, 3, 4, \dots\}$ is the set of natural numbers.
 - $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ is the set of integers (whole numbers with their negatives).
 - $\mathbb{Q}$ is the set of rational numbers, or quotients of integers. Formally:

$$\mathbb{Q} = \left\{ r : r = \frac{p}{q}, p, q \in \mathbb{Z} \right\}$$

- $\mathbb{R}$ is the set of real numbers, or all rational and irrational numbers together.
- ▶ We can use these sets to be more precise in how we write sets formally. For example:

$$Y = \{n \in \mathbb{N} : n \text{ is a multiple of } 11\}$$

would be read aloud as “ Y is the set of *all natural numbers* n such that n is a multiple of 11”.

- ▶ **Question:** If n is a natural number, is n an integer?
- ▶ **Answer:** Yes! The integers contain all positive or negative whole numbers.
- ▶ So every natural number is also an integer. To put it another way, every element of $\mathbb{N}$ is an element of $\mathbb{Z}$.
- ▶ Whenever A and B are sets, and every element of A is an element of B , then we say A is a subset of B , and write $A \subseteq B$.
- ▶ Example: $A = \{14, 49, 77\}$, and $B = \{n \in \mathbb{N} : n \text{ is a multiple of } 7\}$.
Every element of A is a multiple of 7, so every element of A is also in B , so $A \subseteq B$. Or,
$$\{14, 49, 77\} \subseteq \{n \in \mathbb{N} : n \text{ is a multiple of } 7\}.$$
- ▶ Notice that **every set is a subset of itself**: $A \subseteq A$ for every set A .

- ▶ A set can be finite — it can contain a single element, for example. The set $\{1\}$ is a set; it's the set containing 1.
- ▶ **But this is not the same as the number 1.** It's the same thing as the difference between a biscuit, and a box that contains a biscuit.
- ▶ **Subsets** are smaller collections within a set, and are denoted with $\subseteq$. **Elements** are individual members of that set, and are denoted with $\in$:

$$2 \in \mathbb{N}, \quad \{2, 4, 6, 8, \dots\} \subseteq \mathbb{N}, \quad \{2\} \subseteq \mathbb{N}$$

- ▶ The following statements are “grammatically correct”:

$$2 \in \mathbb{N}, \quad \{2\} \subseteq \mathbb{N}$$

and the following are “grammatically incorrect”:

$$\{2\} \in \mathbb{N}, \quad 2 \subseteq \mathbb{N}.$$

- ▶ The notion of “**subsets**” is analogous to the notion of “**less than [or equal]**” for numbers; in a sense, if A and B are sets and $A \subseteq B$, then A is “smaller than” B .
- ▶ When are sets equal?
- ▶ Sets A and B are equal if they have the exact same elements. Or...
- ▶ $A = B$ if every element of A is an element of B , and every element of B is an element of A . Or...
- ▶ $A = B$ if $A \subseteq B$ and $B \subseteq A$.
 - Notice this is starting to look like a compound statement in logic... (which we will be looking at in next part of Week 1).
- ▶ Note (again): $A \subseteq A$ for every subset.
- ▶ A subset of a set that is not equal to the main set is called a **proper subset** (e.g., $\mathbb{N}$ is a proper subset of $\mathbb{Z}$).

Suppose

$$C = \{n \in \mathbb{N} : n \text{ is a multiple of } 6\},$$

$$D = \{n \in \mathbb{N} : n \text{ is a multiple of } 3 \text{ and a multiple of } 2\}.$$

Is it true that $C = D$?

This would mean two things:

- ▶ every element of C is an element of D (i.e., every multiple of 6 is a multiple of both 3 and 2); and,
- ▶ every element of D is an element of C (i.e., every number that's a multiple of both 3 and 2 is also a multiple of 6).

Theorem. $C = D$, where

$$C = \{n \in \mathbb{N} : n \text{ is a multiple of } 6\},$$

$$D = \{n \in \mathbb{N} : n \text{ is a multiple of } 3 \text{ and a multiple of } 2\}.$$

Proof. If $n \in C$, then $n = 6k$ for some $k \in \mathbb{N}$.

Then, $n = 3(2k)$, and $n = 2(3k)$, so n is a multiple of both 3 and 2, so $n \in D$. Therefore $C \subseteq D$.

Now, if $n \in D$, then $n = 2k$ and $n = 3r$ for $k, r \in \mathbb{N}$.

This means $2k = 3r$, so r must be even.

So, $r = 2p$ for some $p \in \mathbb{N}$, so $n = 3r = 3(2p) = 6p$,

so n is a multiple of 6: $n \in C$.

So $D \subseteq C$.



- ▶ There's one more very important set: the so-called **empty set**, $\emptyset = \{\}$.
- ▶ The empty set is the set that contains — *nothing*!
- ▶ It's a surprisingly useful set to consider in abstract set theory, particularly because it is a subset of every set:

$$\emptyset \subseteq A \quad \text{for every set } A.$$

- ▶ If this sounds weird, suppose you have four books in your flat. Those four books form a set.
- ▶ You're carrying some of those books in your bag — a subset of those books.
- ▶ You could have one book, or two, or three, or all four (your entire collection would then be in your bag; the subset of books in your collection that you're carrying is equal to the set of all books in your collection).
- ▶ Or you could have *none* in your bag. Your bag could be *empty*. The subset of books from your collection that you're carrying contains an empty set — no books.

Operation	Notation	Definition
Union	$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$	is a new set that contains all the elements that are in set A, or in set B, or in both (OR)
Intersection	$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$	is a new set that contains all the elements that are common to both set A and set B (AND).
Complement	$A^c \text{ or } \bar{A} = \{x \mid x \in U \text{ and } x \notin A\}$	is the set of all elements that are not in A, but are within a larger, predefined set called the universal set U .
Difference	$A \setminus B = \{x \mid x \in A \text{ and } x \notin B\}$	is a new set that contains all the elements that are in set A but are not in set B.
Symmetric Difference	$A \triangle B = \{x \mid (x \in A \text{ and } x \notin B) \text{ or } (x \notin A \text{ and } x \in B)\}$	is a new set that contains all the elements that are in either set A or B, but not in both (XOR)

- ▶ Consider the sets $A = \{2, 4, 6, 8, 10, 12\}$, and $B = \{3, 6, 9, 12, 15\}$.
- ▶ Consider the following:

$$C = \{n : n \in A \text{ and } n \in B\}, \quad D = \{n : n \in A \text{ or } n \in B\}.$$

- ▶ We give “AND” and “OR” special names in set theory:

 The intersection of sets A and B is the set containing elements in A **and** B :

$$C = A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

 The union is the set containing elements of A **or** B :

$$D = A \cup B = \{x : x \in A \text{ or } x \in B\}.$$

- ▶ In this example,

$$A \cap B = \{6, 12\}, \quad A \cup B = \{2, 3, 4, 6, 8, 10, 12, 15\}$$

- The **set difference** of two sets A and B , is a set, denoted

$$A \setminus B \quad \text{or} \quad A - B,$$

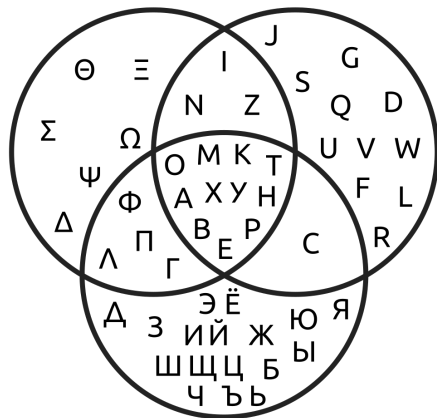
whose elements are those that belong to A , but not to B , that is

$$A \setminus B = \{x \mid x \in A \cap x \notin B\}$$

- The **symmetric difference** of two sets A and B is the set containing
- elements of set A , which are not in B ,
AND
 - elements of set B , which are not in A

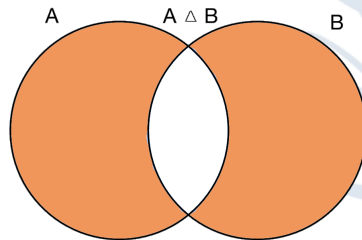
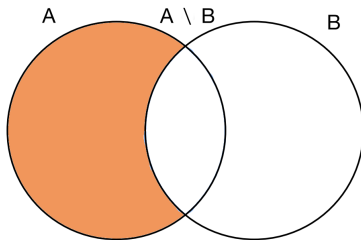
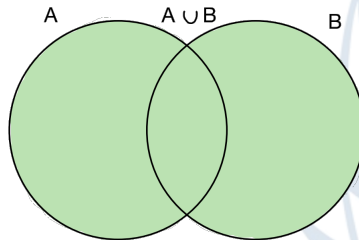
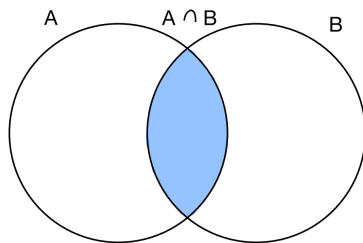
$$A \triangle B = \{x \mid (x \in A \wedge x \notin B) \vee (x \notin A \wedge x \in B)\}$$

A **Venn diagram** is a widely used diagram style that shows the logical relation between sets, popularized by John Venn (1834–1923) in the 1880s.

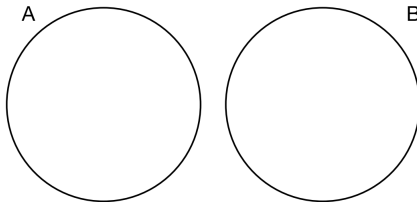


Example: Venn diagram showing the uppercase glyphs shared by the Greek (upper left), Latin (upper right), and Russian Cyrillic (bottom) alphabets [[Wikipedia::Venn diagram](#)]

- It is sometimes useful to consider the intersection, union, difference and symmetric difference of sets by drawing Venn diagrams.



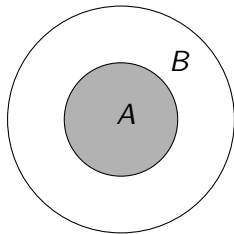
- ▶ Suppose A and B are sets that have no elements in common. Then $A \cap B = \emptyset$. We say that A and B are disjoint.



- ▶ If $A \subseteq B$, then $A \cap B = A$. If set A is a subset of set B , it means every element in A is also present in B . Therefore, the elements that are common to both sets (their intersection) will simply be all the elements of A .
- ▶ $A \cap B = B \cap A$. This is the **commutative property** of intersection. The order in which you intersect two sets does not change the result; the set of elements common to A and B is the same as the set of elements common to B and A .
- ▶ $A \cup B = B \cup A$. This is the **commutative property** of union. Similar to intersection, the order in which you combine the elements of two sets does not matter; the resulting set is the same.
- ▶ $A \cup (B \cup C) = (A \cup B) \cup C$. This is the **associative property** of union. When taking the union of three or more sets, the way they are grouped does not affect the final result. All elements from all three sets will be in the final union.
- ▶ $A \cap (B \cap C) = (A \cap B) \cap C$. This is the **associative property** of intersection. When finding the intersection of three or more sets, you can group them in any way you want and the result will still be the set of elements common to all three.

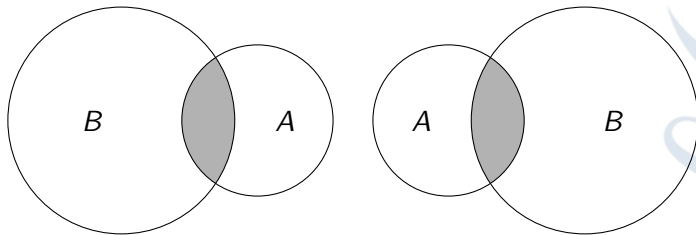
a.k.a. **Absorption Law for Subsets:** $A \subseteq B \Rightarrow A \cap B = A$

If set A is a subset of set B , it means every element in A is also present in B . Therefore, the elements that are common to both sets (their intersection) will simply be all the elements of A .



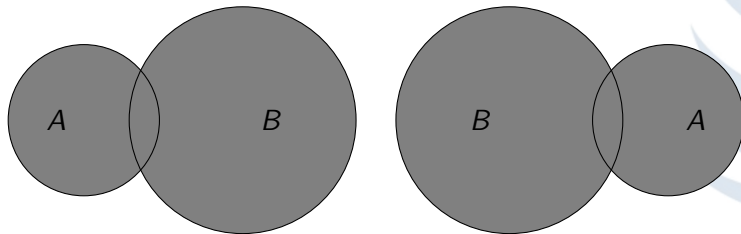
Commutative Property of Intersection: $A \cap B = B \cap A$

This is the **commutative property** of intersection. The order in which you intersect two sets does not change the result; the set of elements common to A and B is the same as the set of elements common to B and A .



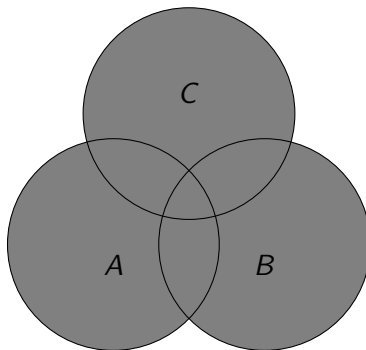
Commutative Property of Union: $A \cup B = B \cup A$

This is the **commutative property** of union. Similar to intersection, the order in which you combine the elements of two sets does not matter; the resulting set is the same.



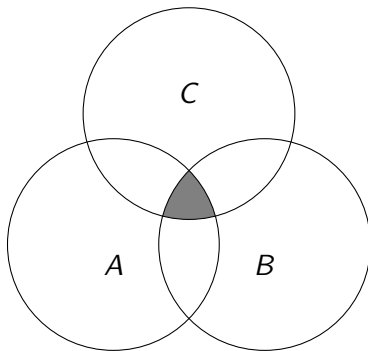
Associative Property of Union: $A \cup (B \cup C) = (A \cup B) \cup C$

This is the **associative property** of union. When taking the union of three or more sets, the way they are grouped does not affect the final result. All elements from all three sets will be in the final union.



Associative Property of Intersection: $A \cap (B \cap C) = (A \cap B) \cap C$

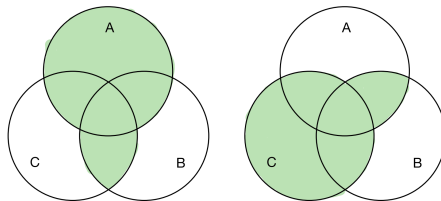
This is the **associative property** of intersection. When finding the intersection of three or more sets, you can group them in any way you want and the result will still be the set of elements common to all three.



Is the following statement true?

For any sets A, B, C , we have $A \cup (B \cap C) = C \cup (B \cap A)$.

► **Solution.** Looking at the Venn diagrams, the answer seems to be no:



- However, **this is not a proof!!** We need to find a concrete counter-example.
- One option: $A = \{a\}$, $B = \{b\}$, and $C = \{c\}$. Then $B \cap C = B \cap A = \emptyset$, but $A \cup (B \cap C) = \{a\} \cup \emptyset = \{a\} \neq \{c\} = \{c\} \cup \emptyset = C \cup (B \cap A)$.

- ▶ A set is a (finite or infinite) collection.
- ▶ The objects in a set are elements, written as $x \in A$ to mean “ x is an element of the set A ”.
- ▶ A subset is a collection that is contained in a larger set. This collection may be empty, or it may be the entire set.
- ▶ Two sets A and B are equal if $A \subseteq B$ and $B \subseteq A$.
- ▶ The union of sets $A \cup B$ is the set of points contained in one of A or B (or both).
- ▶ The intersection of sets $A \cap B$ is the set of points contained in both A and B .
Two sets are disjoint if $A \cap B = \emptyset$.
- ▶ You can study complex intersections and unions by looking at Venn diagrams.

Quiz! – <https://coventryassessment.inspera.com>

